

Lecture 8: Model Selection for Classification

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10 ECTS Credits

Department of Information and Communication Technology

Faculty of Engineering and Science

Recap

- Error Estimation Rules
- Error Estimation Performance:
 - Deviation Distribution
 - Bias, Variance, RMS, Tail Probabilities
 - Consistency
- Test-Set Error Estimation
- Resubstitution
- Cross-Validation



Model Selection for Classification

- Trying to answer a couple of questions:
 - Which classification rule should one choose given a problem?
 - How to pick the free parameters of a classification rule?
- These questions can be seen as model selection questions.
- Answer depends on the ratio of complexity to sample size.
- Complexity of a problem is defined by:
 - degrees of freedom in classification rule, and
 - dimensionality of the problem.
- Model selection procedures seek classifiers that fit the training data well while exhibiting a good complexity to sample size ratio.

Classification Complexity

- Let $\mathcal{C}$ be a **classifier space**: an arbitrary set of classifiers.
- A classification rule Ψ_n picks a classifier $\psi_n \in \mathcal{C}$ based on the sample S_n .
- E.g.
 - LDA, linear SVM, and perceptron classification rules produce classifiers in the space $\mathcal{C}$ of linear classifiers.
 - QDA classification rule produces classifiers in the space $\mathcal{C}'$ of quadratic classifiers
 - Here, $\mathcal{C} \subset \mathcal{C}'$
 - More complex classification rule results in larger $\mathcal{C}$.
- Parameters like weights of a fixed-architecture NN must be estimated from the training data: modeled as part of $\mathcal{C}$.
- Parameters such as number of hidden layers, neurons in an NN are not directly estimated from the training data and hence called **free parameters** and are assumed fixed.
- Thus, different free parameters lead to different classifier space $\mathcal{C}$.

Classification Complexity

- Based on classification error, the best classifier in $\mathcal{C}$ can be defined as:

$$\psi_{\mathcal{C}} = \arg \min_{\psi \in \mathcal{C}} \varepsilon[\psi] = \arg \min_{\psi \in \mathcal{C}} P[\psi(X) \neq Y]$$

with $\varepsilon_{\mathcal{C}} = \varepsilon[\psi_{\mathcal{C}}]$ (Best-in-class error).

- Given S_n , the designed classifier is $\psi_{n,\mathcal{C}} = \Psi_n(S_n) \in \mathcal{C}$ with error $\varepsilon_{n,\mathcal{C}} = \varepsilon[\psi_{n,\mathcal{C}}]$
- The **approximation error** is the difference between the best error in the classifier space and the Bayes Error:

$$\Delta_{\mathcal{C}} = \varepsilon_{\mathcal{C}} - \varepsilon^*$$

- The approximation error shows how well the best classifier in the hypothesis class $\mathcal{C}$ can approximate the performance of the Bayes classifier.

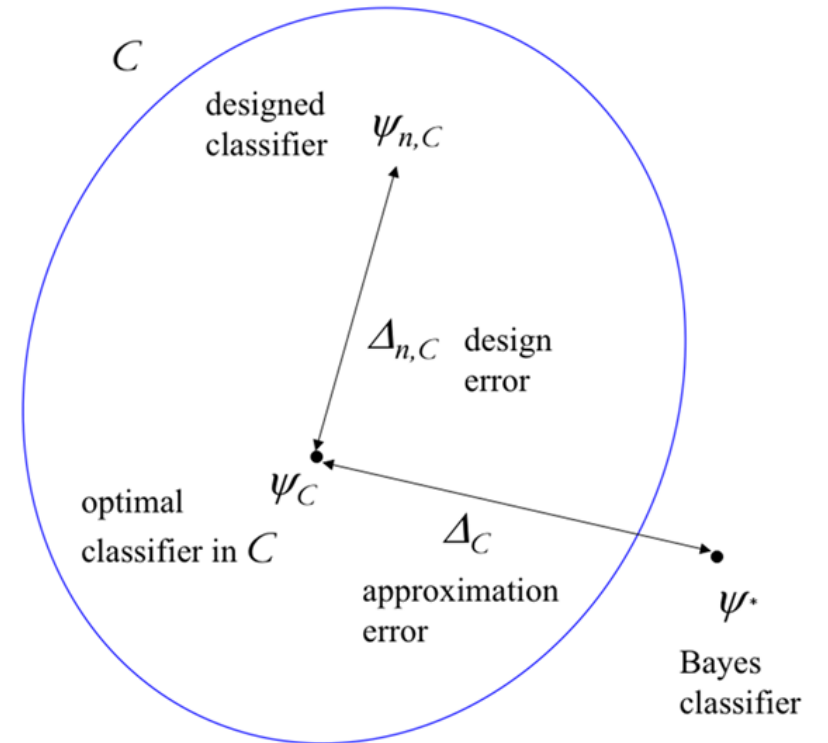
Classification Complexity

- The **design error** is the difference between the error of the designed classifier and the best-in-class error:

$$\Delta_{n,\mathcal{C}} = \varepsilon_{n,\mathcal{C}} - \varepsilon_{\mathcal{C}}$$

- Design error reflects how close to the best possible classifier in $\mathcal{C}$ is the classifier designed based on the available data.
- Thus, we can decompose the error of the designed classifier as:

$$\varepsilon_{n,\mathcal{C}} = \varepsilon^* + \Delta_{\mathcal{C}} + \Delta_{n,\mathcal{C}}$$

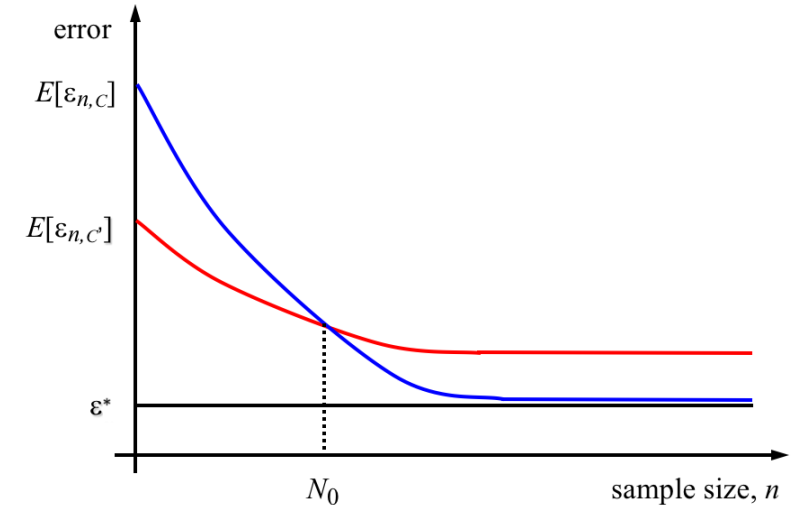


Classification Complexity

- The expected classification error gives an acceptable performance measure for a classification rule.
- Applying expectation on both sides:
$$E[\varepsilon_{n,\mathcal{C}}] = \varepsilon^* + \Delta_{\mathcal{C}} + E[\Delta_{n,\mathcal{C}}]$$
- To pick the best $\mathcal{C}$, both $\Delta_{\mathcal{C}}$ and $E[\Delta_{n,\mathcal{C}}]$ should be as small as possible.
- A more complex classification rule (larger $\mathcal{C}$) yields smaller approximation error $\Delta_{\mathcal{C}}$.
- If we eventually make $\mathcal{C}$ big enough, Bayes classifier will be in $\mathcal{C}$ and hence $\Delta_{\mathcal{C}} = 0$.
- However:
 - Smaller $\mathcal{C} \Rightarrow$ smaller $E[\Delta_{n,\mathcal{C}}]$ but larger $\Delta_{\mathcal{C}}$.
 - Larger $\mathcal{C} \Rightarrow$ smaller $\Delta_{\mathcal{C}}$ but larger $E[\Delta_{n,\mathcal{C}}]$.

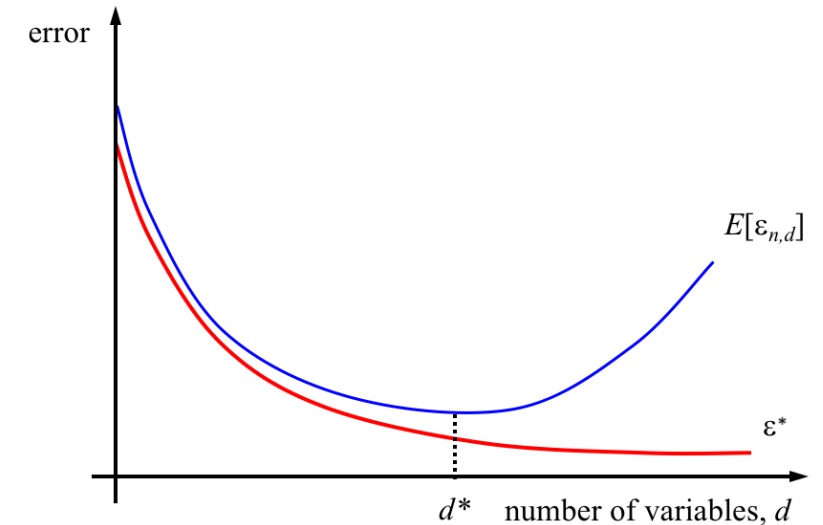
Classification Complexity

- This **complexity dilemma** is a reflection of the bias-variance dilemma: One can generally control bias or variance, but not both.
- The optimal operation point in the complexity dilemma is determined by the sample size.
- For large n , $E[\Delta_{n,\mathcal{C}}]$ tends to be small. i.e., the objective is to minimize $\Delta_{\mathcal{C}}$, we can use complex classification rule.
- For small n , $E[\Delta_{n,\mathcal{C}}]$ tends to be large and minimizing $\Delta_{\mathcal{C}}$ becomes a secondary objective, we avoid using complex classification rules.
- The relationship between complexity and sample size can also be seen in the **scissors plot**.



Classification Complexity

- Increasing dimensionality d has the effect of increasing the size of $\mathcal{C} \rightarrow$ increases classification complexity.
- For fixed sample size, n , the difference:
$$E[\varepsilon_{n,d}] - \varepsilon_d^* = \Delta_d + E[\Delta_{n,d}]$$
- The above difference increases with d , as $E[\Delta_{n,d}]$ cannot be controlled.
- $E[\varepsilon_{n,d}]$ decreases following the decrease in ε^* but increases eventually.
- This phenomenon is called **peaking phenomenon** or **Hughes phenomenon**, or **curse of dimensionality**.
- Thus, feature selection is a type of model selection, which would result in an optimal dimensionality d^* .
- d^* is usually unknown and is dependent on the distribution of the problem.



Model Selection Methods

- Assume that classification rule $\Psi_{n,k}$ uses training data S_n to pick a classifier $\psi_{n,k}$ in a space of classifiers $\mathcal{C}_k$, for $k = 1, 2, \dots, N$.
- We assume finite number of spaces of classifiers, N for simplicity:
 - The choice is between a small number of different classification rules,
 - Any continuous free parameters can be discretized into a finite grid.
- The goal of model selection is to pick a classifier $\psi_n = \psi_{n,k_n} \in \mathcal{C}_{k_n}$ that has the smallest error among the candidates.
- The selected classification rule is then Ψ_{n,k_n} .
- Model selection is about optimization in the available space of classifiers.

Validation Error Minimization

- Selected classifier ψ_n is the one that produces smallest error on an independent sample $S_m = \{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_m, Y_m)\}$, known as validation set.

$$\psi_n = \arg \min_{\psi_{n,k}} \hat{\varepsilon}[\psi_{n,k}] = \arg \min_{\psi_{n,k}} \frac{1}{m} \sum_{i=1}^m |Y_i - \psi_{n,k}(\mathbf{X}_i)|$$

- What we really want is a classifier ψ_n^* that achieves the smallest true classification error:

$$\psi_n^* = \arg \min_{\psi_{n,k}} \varepsilon[\psi_{n,k}] = \arg \min_{\psi_{n,k}} E[|Y_i - \psi_{n,k}(\mathbf{X})|]$$

- We care about the difference between the true error $\varepsilon[\psi_n]$ of the sample-based classifier and the true error $\varepsilon[\psi_n^*]$ of the best classifier: $\varepsilon[\psi_n] - \varepsilon[\psi_n^*]$

Validation Error Minimization

- By definition of Bayes classifier $\varepsilon[\psi_n^*] = \min_k \varepsilon[\psi_{n,k}]$:
$$\varepsilon[\psi_n] - \varepsilon[\psi_n^*] = \varepsilon[\psi_n] - \min_k \varepsilon[\psi_{n,k}] = \varepsilon[\psi_n] - \hat{\varepsilon}[\psi_n] + \hat{\varepsilon}[\psi_n] - \min_k \varepsilon[\psi_{n,k}]$$
- By definition of how ψ_n was selected $\hat{\varepsilon}[\psi_n] = \min_k \hat{\varepsilon}[\psi_{n,k}]$:
$$\varepsilon[\psi_n] - \varepsilon[\psi_n^*] = \varepsilon[\psi_n] - \hat{\varepsilon}[\psi_n] + \min_k \hat{\varepsilon}[\psi_{n,k}] - \min_k \varepsilon[\psi_{n,k}]$$
- By the property $\min_k a_k - \min_k b_k \leq \max_k (a_k - b_k) \leq \max_k |a_k - b_k|$:
$$\varepsilon[\psi_n] - \varepsilon[\psi_n^*] \leq \varepsilon[\psi_n] - \hat{\varepsilon}[\psi_n] + \max_k |\hat{\varepsilon}[\psi_{n,k}] - \varepsilon[\psi_{n,k}]|$$
- The maximum absolute difference across the errors of all classifiers must be at least as large as the absolute difference for any specific classifier:
$$|\hat{\varepsilon}[\psi_n] - \varepsilon[\psi_n]| \leq \max_k |\hat{\varepsilon}[\psi_{n,k}] - \varepsilon[\psi_{n,k}]|$$

Validation Error Minimization

- Now,

$$\varepsilon[\psi_n] - \hat{\varepsilon}[\psi_n] \leq |\hat{\varepsilon}[\psi_n] - \varepsilon[\psi_n]| \leq \max_k |\hat{\varepsilon}[\psi_{n,k}] - \varepsilon[\psi_{n,k}]|$$

- Thus,

$$\varepsilon[\psi_n] - \varepsilon[\psi_n^*] \leq \varepsilon[\psi_n] - \hat{\varepsilon}[\psi_n] + \max_k |\hat{\varepsilon}[\psi_{n,k}] - \varepsilon[\psi_{n,k}]|$$

implies:

$$\varepsilon[\psi_n] - \varepsilon[\psi_n^*] \leq \max_k |\hat{\varepsilon}[\psi_{n,k}] - \varepsilon[\psi_{n,k}]| + \max_k |\hat{\varepsilon}[\psi_{n,k}] - \varepsilon[\psi_{n,k}]|$$

or

$$\varepsilon[\psi_n] - \varepsilon[\psi_n^*] \leq 2 \max_k |\hat{\varepsilon}[\psi_{n,k}] - \varepsilon[\psi_{n,k}]|$$

Validation Error Minimization

- Irrespective of the feature-label distribution, $\max_k |\varepsilon[\psi_{n,k}] - \hat{\varepsilon}[\psi_{n,k}]|$ is bounded by (proved using Union Bound and Hoeffding's Inequality):

$$P\left(\max_k |\varepsilon[\psi_{n,k}] - \hat{\varepsilon}[\psi_{n,k}]| > \tau\right) \leq 2Ne^{-2m\tau^2}, \quad \forall \tau > 0$$

- Theorem:** If a validation set with m points is used to select a classifier among N choices, the difference in true error between the selected classifier ψ_n and the best classifier ψ_n^* satisfies

$$P(\varepsilon[\psi_n] - \varepsilon[\psi_n^*] > \tau) \leq 2Ne^{-\frac{m\tau^2}{2}}, \quad \forall \tau > 0$$

- As the size of validation set m approaches infinity, we are certain to make the right selection.

Validation Error Minimization

- When m is larger compared to N , the bound becomes narrower.
- Intuitively, it is easier to select among few choices than many choices.
- $\hat{\varepsilon}[\psi_n]$ is not an unbiased estimator of $\varepsilon[\psi_n]$ since the validation set is not independent of ψ_n .
- $\hat{\varepsilon}[\psi_n]$ is likely to be optimistically biased as an estimator of $\varepsilon[\psi_n]$.
- When we select a model based on validation performance, we're using the validation data for evaluation and selection.
- This results in the optimistic bias leading to overconfidence in model performance, which could result in potential disappointment when deployed in real world settings.

Training-Validation-Testing Strategy

- To have an unbiased estimator of $\varepsilon[\psi_n]$, we can use a test set independent of both the training and validation sets.
- This leads to a three-way split of the data into a training, validation, and testing sets.
- The classifiers are designed on the training data under different models.
- The validation data is used to select the best model.
- The performance of the selected classifier is then assessed on the test data, which stands in for real-world performance.
- E.g. This strategy is used to select the number of epochs an NN is trained:
 - The training is stopped once the validation error is not improving for a preset number of epoch.
 - The test set is not used to make this decision to keep test-set error estimator unbiased.

Training Error Minimization

- When data is scarce, validation error minimization approach is unrealistic.
- In this case, we need to use as much available data as possible for training.
- In this case, we select a data-efficient error estimator $\hat{\varepsilon}_n$ like resubstitution and pick a classifier that produces smallest error estimate:

$$\psi_n = \arg \min_{\psi_{n,k}} \hat{\varepsilon}_n[\psi_{n,k}]$$

- Again, we are interested in the classifier ψ_n^* that achieves the smallest true classification error:

$$\psi_n^* = \arg \min_{\psi_{n,k}} \varepsilon[\psi_{n,k}]$$

- The inequality still holds:

$$\varepsilon[\psi_n] - \varepsilon[\psi_n^*] \leq 2 \max_k |\hat{\varepsilon}[\psi_{n,k}] - \varepsilon[\psi_{n,k}]|$$

Training Error Minimization

$$\varepsilon[\psi_n] - \varepsilon[\psi_n^*] \leq 2 \max_k |\hat{\varepsilon}_n[\psi_{n,k}] - \varepsilon[\psi_{n,k}]|$$

- If the error estimator $\hat{\varepsilon}_n$ can estimate the error uniformly well over the set of all classifiers $\{\psi_{n,k} \mid k = 1, \dots, N\}$, the RHS bound is small.
- This means that good model selection performance can be guaranteed. However, theoretically, this is not the case.
- A popular approach is to use cross validation to pick the value of a free parameter in a grid search, due to the small bias.
- The cross-validation grid search works in practice, if the sample size is not too small.

Thank You